

An Edge-based Mesh Segmentation Method for Engineering Objects

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Abstract—This paper proposes a triangle mesh segmentation technique for conventional engineering objects. We exploited two key aspects of engineering geometry: (1) the intrinsic discontinuities existing in boundary surfaces and (2) the anisotropic behavior of true discontinuities. A two run non-linear diffusion algorithm based on an optimal estimation theory is designed to filter out undesired discontinuities. The first run is for treating C1 continuities where the crease edges are detected using anisotropic re-estimation of triangle normal. The second run is for handling C2 continuities in which transition edges are derived through diffusion of curvature tensor associated with each triangle. Morphological operator and geometric snake model are implemented for interpolating smooth and closed feature loops. Results indicate that our method can determine surface patch layout satisfactorily.

Keywords- mesh segmentation, reverse engineering

I. INTRODUCTION

Mesher, although not a mainstream representation in CAD systems, are frequently generated and exported to downstream applications such as Finite Element Analysis (FEA), Rapid Prototyping (RP), and Computer Aided manufacturing (CAM). Besides, meshes can also originate from reverse engineering tasks. Mesh segmentation is a fundamental problem for many mesh based applications. Some of these applications require decomposition to higher level sub-components. Examples include animation and morphing [1-2], shape matching [3]. Other applications, such as texture mapping [4], remeshing and simplification [5], reverse engineering [6-9], require lower level sub-surfaces. In this work, we studied the problem of mesh segmentation for engineering objects, targeting mainly the application of reverse engineering. The objective is to group the unstructured data: triangles into surface patches, while each of the patches logically belongs to a single primitive surface.

There is a large body of research dealing with mesh segmentation, general surveys can be found in [10-11]. Ref. [9] is a good source for the introduction on segmentation and surface fitting techniques for reverse engineering. In general, previous approaches form two very different categories: (1) Edge-based approaches work by first identifying features edges according to surface boundaries and then interpolating them to form smooth boundaries [12-14]. Feature regions are extracted through examining differential properties using different orders. The difficulties of edge-based methods are the precise estimation of differential properties for noisy models and the locating of smooth and closed feature loops. (2) Region-based approaches take the dual path to avoid the difficult edge

localization problem. The idea is to group vertices/triangles into regions corresponding with the same subjacent primitive surfaces first, and the boundaries being derived after [5, 15-18]. Region-based methods count on suitable seeds being selected beforehand, and heavy post processing is usually required for boundary rectification.

In this paper, we developed an edge-based segmentation method which is mainly applied to engineering objects. The method starts with the per-face local geometry estimation. It further locates the surface discontinuities by adaptively filtering out the unwanted noise through two-run non linear diffusion processes. The anisotropic filtering technique developed in our previous work [19] was used for the differential parameters estimation on noisy models. The left-over sharp changes of each run, corresponding to the first order (C1) discontinuities and second order (C2) discontinuities are taken as loci of true features. These sharp changes are then carefully traced using morphologic operator [12] and geometric snake [14] to form closed feature loops. The final segmentation is achieved by a region growing process inside each of the feature loops.

II. PER-FACE ESTIMATOR AND OPTIMAL ESTIMATION

As the conventional per-vertex local geometry estimation method favors the region-based segmentation, we used a simple per-face discrete differential parameter estimator. It benefits our edge based method since the sharp change of differential properties between two neighboring faces naturally defines a potential feature edge.

For each triangle in a mesh, its normal direction has already been defined or can be calculated easily; its curvature (as a 3x3 tensor) can be estimated initially based on the normal variations of its three vertices using least square fitting [19]. These initial estimations are error-prone because of two types of discontinuities existing on scanned data: 1) additive uncorrelated random noise and, 2) discontinuities on the underlying surfaces. In both cases, normals are not well defined along the discontinuous regions, thus the curvature tensors estimated in the neighboring triangles are unstable as well. We adapted the optimal estimation theory into a two run non-linear diffusion process, and decoupled the smoothing and denoising problem in two stages, one for normal consistency and the other for curvature consistency. Each stage preserves the anisotropic discontinuities while removing the isotropic noises.

The main idea of the optimal estimation theory [20] is that, given a noisy source, its contribution to predict the behavior of

a target in the next iteration is dependent on the variance of its prediction error in the past. A source with a lower variance of estimation error is considered to have better fitness and will be given higher weight in the next prediction. This mechanism works elegantly for the purpose of anisotropic filtering on initially estimated surface parameters. The normal or curvature tensor, if both denoted as $\Delta f^{(k+1)}$ for a triangle f at a new iteration $k+1$, is estimated by the weighted average of the differential properties of its neighbors in $N(f)$,

$$\Delta f^{k+1} = \frac{\sum_{s \in N(f)} w_s^k \Delta_s^k}{\sum_{s \in N(f)} w_s^k} \quad (1)$$

where w_s^k is the weight assigned to a neighbor s for measuring how much the prediction of s will contribute to the behavior of the target f . The weight is assigned based on the optimal estimation theory formulated as follows.

$$w_s^k = \exp\left(-\frac{\sigma_s^{2k}}{\gamma^k}\right) \quad (2)$$

$$\sigma_s^{2k} = \frac{1}{k} \sum_{i=0}^k \epsilon_s^{2i} \quad (3)$$

$$\gamma^k = \frac{2}{|N(f)|} \sum_{s \in N(f)} \sigma_s^{2k} \quad (4)$$

Where σ_s^{2k} is the variance of prediction error in k th iteration and γ^k is the smoothing control parameter which is set by the mean of the error variance among all the triangles in $N(f)$. ϵ_s is the prediction error, i.e. the difference between Δs and Δf .

Using Equations (1) to (4), we applied the anisotropic filtering first on the pre-estimated triangle normals and then on the curvature tensors, both preserving the desired feature edges. In each run, the setting for the error measurement is different, we skip the detail here, readers can found in [20]. After these two run filtering process, most of the sharp changes in terms of triangle normal directions or curvatures will behave in an anisotropic fashion, which means they are presenting more in a thin line style.

III. EDGE DETECTION, TRACING AND SEGMENTATION

Two types of features are identified during the edge detection stage: crease edges and transition edges, corresponding to $C1$ discontinuities and $C2$ discontinuities.

Crease edges tie up with sharp normal transitions or $C1$ discontinuities. We obtain them by thresholding the dihedral angles. Edges with dihedral angle larger than a threshold T_n are marked as sharp edges. Triangles with $\max(|k_{min}|, |k_{max}|)$ (k_{min} and k_{max} are the two principal curvatures) larger than a value T_k are marked with feature triangles, with their vertices marked as feature vertices. Since sharp edges manifest themselves as a high curvature region, all sharp vertices are also tagged as feature vertices. Our idea for crease edge detection is to thinning the feature region to get a thin-line skeleton, which is achieved through the morphological “skeletonize” operator adapted from [12]. Then the open edges are closed into loops using the geometric snake tool [14].

By using the morphological skeletonize operator, the outmost layer is removed iteratively till a thin-line skeleton is achieved, while this thinning is done without changing the topology of the feature regions. The algorithm for “Morphological skeletonize” operator is listed below in Algorithm 1. Note that a feature point is marked as key point if the total number of variations among the sequence of neighboring vertices ordered clockwise is ≥ 4 (a variation means a change from a feature point to a non-feature point or vice versa.)

Algorithm 1 Morphological skeletonize

Input: A Mesh, Feature Point list FPLST

Output: skeletonized FPLST

1: initialize two empty vertex list CENLST, BONDST

2: **For** each vertex v in FPLST

3: **if** all neighbors of v are all in FPLST

4: add v to CENLST, add neighbors to BONDST

5: **For** each vertex p in BONDST

6: **if** p is NOT in CENLST, AND p is NOT key point

7: remove p from FPLST

8: remove p from BONDST.

9: Recursively apply step 1-8 till size of FPLST doesn't change

10: remove all non-key vertex in FPLST

After applying the “Morphological skeletonize” operator, the feature points will be appeared as thin-line set of points. But chances are that they may not form closed loops. If this is the case, gaps will be filled using the geometric snake algorithm. Our snake model works as follows. All the sharp edges with only one sharp neighboring edge are labeled as Open Snake Edge (OSE). For each OSE, its non-sharp edge neighbors are considered as Potential Snake Edge (PSE). The snake with OSE as its head, always slither to a PSE which has lowest weight p until it ends up with closed loop. For each PSE edge e_i , its weight p_i is defined by:

$$p_i = \omega_\theta \times \theta(e_i, e_{OSE}) + F_s(e_i) \quad (5)$$

The First item is concerned with the smoothness of snake. Where $\theta(e_i, e_{OSE}) \in [0, 180]$ is the angle between the edge OSE and a PSE e_i . The latter terms captures the feature energy along an edge which is defined as:

$$F_s(e_i) = \omega_e \times D_E(e_i) + \omega_k \times (D_k(e_i)) \quad (6)$$

The first item in Eq. (6) depends on the angles between PSE and the principle directions of two faces that sharing a PSE (see Fig. 1):

$$D_E(e_i) = 90 + \min(\theta_{\min_{i1}}, \theta_{\min_{i2}}) - \theta_{\min_{i12}}$$

The second item in Eq. (6) measures the curvature value variation between neighboring triangles:

$$D_k(e_i) = \frac{\|K \min_{i1} - K \min_{i2}\| + \|K \max_{i1} - K \max_{i2}\|}{\max(K \min_{i1}, K \min_{i2}) + \max(K \max_{i1}, K \max_{i2})}$$

We normalized the value of $D_k(e_i)$ in $[0, 180]$ to keep the coherency with the other two angle scores. ω_θ , ω_n and ω_k are weights for balancing each item's contribution to this energy minimization process.

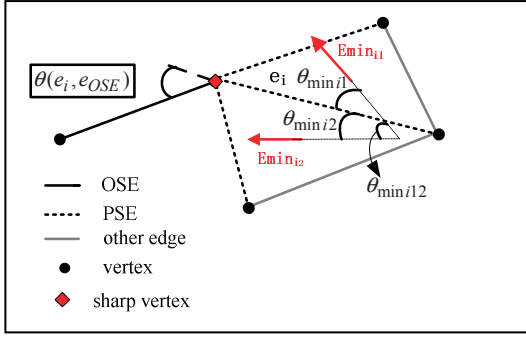


Figure 1. Elements taken into account for the calculation of the energy function.

Through morphological skeleton and snake based tracing, crease edges will all appear in closed feature loops. The first run segmentation can be easily obtained though region growing among these closed feature loops.

Once an initial segmentation is obtained in the first run, all the per-face curvature estimating will be recalculated. Vertex normals along boundaries will give different directions when per-face curvature tensors are estimated. Vertex normal estimation is dependent on which side of the surface is under consideration. The neighborhoods, both for normal estimation, and for the optimal estimation processes, are all composed of only triangles belonging to the same patch region. This prohibits our local geometry estimation being estimated cross the discontinuities captured in crease edges.

In the second run we are interested on the transition edges along which the surfaces are not necessarily bent sharply but the curvatures change in a sharp way. We locate this type of feature initially by thresholding the transition strength S_t of each edge inside a patch. Given the edge e , and the curvature tensors H_{f1} and H_{f2} of its two triangle neighbors: f_1 and f_2 , transition strength S_t is defined as :

$$s_t = \sqrt{\|H'_{f1} - H_{f2}\|^2 + \|H'_{f2} - H_{f1}\|^2} \quad (7)$$

where H'_{f1} is the rotated tensor from H_{f1} to H_{f2} , and H'_{f2} is the rotated tensor from H_{f2} to H_{f1} . (The definition of rotated tensor is explained in [19].)

Considering the vicinity of crease edges may well present a sharp curvature transition, those are not our desired features, so we erode the boundary of each patch inside by n -neighborhood ($n = 3$ for example) of boundary edge and the edge strength S_t will be only estimated in the inside regions. For simplicity in this work, we use a mean filter on edge strength to separate an edge as feature edges or not. Vertices of features edges will be labeled as feature point. We use the same ‘‘Morphological skeletonize’’ algorithm discussed previously to form a thin-line feature edge. All these feature edges, naturally present in closed loop using the skeletonize algorithm, define our final segmentation boundaries.

Notice that the ‘‘Morphological skeletonize’’ operator is sensitive to the threshold setting for feature regions. A small threshold may cause separate but close features intersecting each other and therefore introduces undesired topologies in final skeletons. Large threshold may cause gaps in the

continuous feature regions thus losing the trace of feature lines. To preserve features as much as possible, we tend to favor a small threshold and use post processing to remove undesired boundaries. In this work, we implemented a simple post processor which merges small patches with one of its neighbors. A patch is considered as small if its area ratio to area of the largest patch is blow a small value. A small patch will be merged into a neighbor, with the boundary to that neighbor having the smallest average feature strength.

IV. EXPERIMENTS AND DISCUSSION

In this section, we show the results obtained by above edge detection and mesh segmentation algorithm. Limited by the space, only one example can be shown here. Fig. 2 shows the process of denoising, edge localization and segmentation of a fandisk model (with 23964 triangles, 11984 vertices, 35946 edges). Noises on this model mainly show up near feature regions (Fig 2.a). Due to these noises, initially located sharp edges, although globally following the feature regions (Fig 2.b), are locally irregular and jaggy if one zooms in to Fig 2.e. After 5 iterations of optimal estimation for normal, sharp edges behave more regularly and anisotropically. (See Fig. 2.c and 2.e). After ‘‘skeletonizing’’ the high maximum curvature region (Fig. 2f), open contours still exist (Fig. 2g). We then trace them, using the geometric snake model to close the loops (Fig 2h). First run segmentation result is shown in (fig 2i). In second run of optimal estimation, we did 5 iterations for curvature tensor diffusion inside each patch, after that, sharp transitions among principal directions are identified (Fig 2j) and they are skeletonized as transition edges (Fig. 2k). These edges will be added to the boundaries and guide the final stage of mesh segmentation (see result in Fig 2l). Other results which are not shown here indicate that our method can determine surface patch layout satisfactorily.

Several directions are possible for future work. We need a better filter for locating feature regions before applying ‘‘skeletonize’’ operator. In this paper, we simply specified those as user controlled parameters. Also the number of iterations in each run diffusion, currently set as user defined parameter too, can be controlled by tracking the convergence properties. Also our method has problems when two feature loops are very close. They will either be smoothed out during the optimal estimation process, or connected during the morphological skeletonize process. A careful remeshing or subdivision along these regions may benefit a lot.

V. CONCLUSIONS

In this paper, we presented an algorithm for segmenting noisy meshes into sub patches based on edge localization. In order to find out the feature edges, we separate them to crease and transition edges. Previous treatments of segmentation have ignored the C2 type discontinuities which we detect after non-linear diffusions over the curvature tensor. Without prior knowledge about noise distribution, our method can detect the anisotropic discontinuities while filtering out the isotropic ones. Discontinuities are traced into closed loops to define boundaries of segments. This prepares the surface patches for further classification and fitting tasks in reverse engineering.

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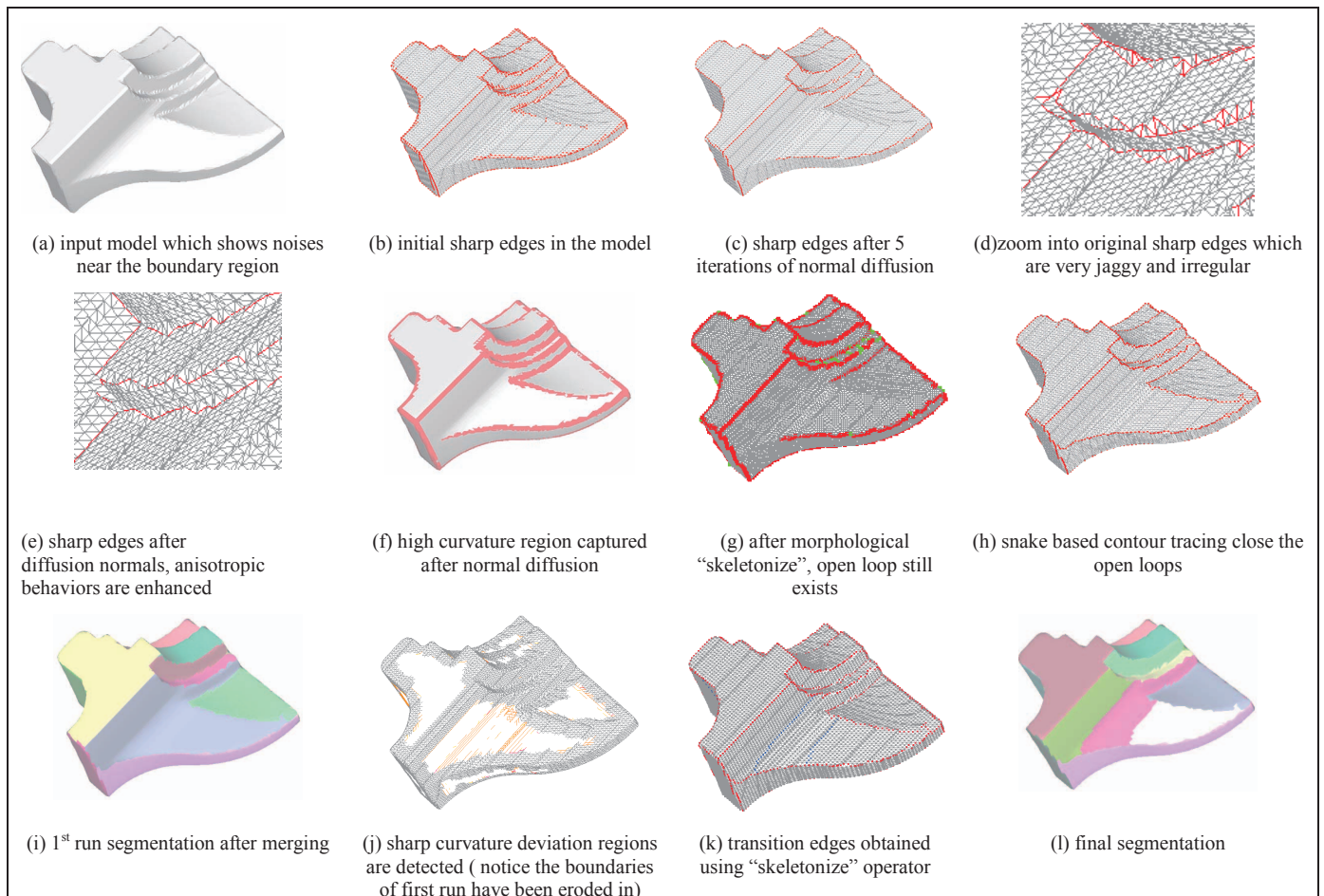


Figure 2. Edge detection and mesh segmentation of a noisy fan disk model.